M1: Approach

Team Number: **191**

Problem Statement : Automatic Category Suggestions for Wikimedia Commons Images

Problem Statement and Our Understanding

Wikimedia Commons is the free media repository used by Wikipedia and its sister projects. Files are organized and found through categories, which volunteers assign by hand. Every day,tons of new files are uploaded, and many end up poorly categorized or not categorized at all. An uncategorized file is difficult to find, search, and reuse, which undermines the core mission of Commons.

The task is to build a system that automatically suggests relevant categories for newly uploaded images. These suggestions assist human editors, who remain in control and approve or discard each one. The system must handle a wide range of subject matter (objects, scenes, events, abstract concepts), generalize to unseen images, remain affordable at Commons scale, tolerate poor-quality photos, and improve as new data arrives.

The category space on Commons is enormous and constantly growing—a fixed output classifier would need re-training every time a new category is created. An image can legitimately belong to multiple categories. The distribution of categories follows a long-tail pattern. Stored labels are often incomplete, and editors tend to file under the most specific category only, leaving out broader parents. Finally, near-identical images from the same uploader can leak between training and test sets, inflating scores. Our approach must directly address each of these challenges.

Our Idea: Treat Categories as Things to Retrieve

Rather than predicting from a fixed list, we convert every image into a compact numerical description (an embedding) using a pre-trained vision-language model from the CLIP family, which places pictures and text in a shared space. Categories are then something we look up and rank using complementary signals whose strengths and weaknesses differ.

Pipeline Overview

`New Image` $\rightarrow$ `CLIP Embedding` $\rightarrow$ `Three Scorers` $\rightarrow$ `Fusion` $\rightarrow$ `Tree Clean-up` $\rightarrow$ `Top Suggestions`

Stage 1: Image Embedding

Each uploaded image is converted into a 512-dimension vector using CLIP and normalized. This gives us one cheap, reusable representation for every subsequent step. The encoder resizes every input to a fixed size, so the pipeline accepts any resolution.

Stage 2: Three Scorers

We use three independent scoring mechanisms, each with different strengths:

- **Zero-Shot Scorer:** We embed the text "a photo of [category name]" for every category using CLIP's text encoder. An image scores high for a category if its embedding is close to the name's embedding. This requires no training and works even for categories with very few or zero examples—our primary tool for the long tail.
- **k-Nearest Neighbour (kNN) Scorer:** For a new image, we find the K most similar training images and copy their categories, giving closer images a bigger vote. This is intuitive, requires no training, allows new images to be added instantly, and provides a built-in explanation by showing the most similar known images.
- **Trained Classifier Head:** A small, lightweight logistic regression model trained on top of the CLIP embeddings. It learns subtle patterns from our dataset. To handle the massive number of negative examples and incomplete labels, we use an asymmetric loss function. This prevents the model from being overwhelmed by negative signals and tolerates near-misses where a correct label was simply never recorded by a volunteer.

Stage 3: Fusion

A logistic regression meta-classifier learns how much to trust each of the three scorers based on performance on a validation set. This is called stacking, and it produces one combined score per category.

Stage 4: Hierarchy Clean-up

We use the Wikimedia category tree to remove redundant suggestions. Following the "most-specific" rule that editors use, if both a specific category (e.g., "Siamese cats") and its parent ("Cats") are suggested, the parent is removed.

Planned Testing Methodology

We will judge the system the way it will really be used: on images uploaded after the ones it learned from.

- **Data Split:** A strict time-based split—the oldest 70% of images for training, the next 15% for validation and tuning the fusion step, and the newest 15% for the final test. This mimics reality and prevents data leakage. We will also remove near-duplicate images from validation and test sets if they are almost identical to a training image.
- **Metrics:** Precision, Recall, and F1-score at $k = 1, 3, 5, 10$, averaged per image. More suggestions raise recall but lower precision—this trade-off is normal and we will present it clearly.
- **Frequency Tiers:** Recall reported separately for common, middle, and rare categories (head, torso, and tail). This will show whether the long tail is genuinely covered or if the model only performs well on popular categories.
- **Category Types:** Results broken down by objects, scenes, events, and abstract topics to ensure the model is not biased towards easily identifiable objects.
- **Robustness:** We will simulate three kinds of damage on the test images—strong blur, shrinking then enlarging, and heavy JPEG compression—and repeat the evaluation.
- **Strict vs. Reviewed Score:** Because stored labels are incomplete, a suggestion counted as wrong may actually be correct. We will manually inspect a sample of "incorrect" suggestions and report both the strict score and a reviewed score. We will never present only the more flattering number.

Expected Outcomes

We expect to deliver a working prototype that demonstrates the feasibility of our retrieval-based approach. The system is designed to be scalable, robust (tested against image degradation), and continually improving (accepted suggestions join the search index immediately with no retraining required). The final deliverable will include a web-based demo where a user uploads an image, receives ranked suggestions with scores, sees the six most similar known images as an explanation, and can accept or reject each suggestion.